

Notes: Kacperczyk, Sialm, and Zheng (JF, 2005)

On the Industry Concentration of Actively Managed Equity Mutual Funds

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 Data sources and sample construction
- 2.2 Industry groups
- 2.3 Industry Concentration Index
- 2.4 Fund performance measures
- 2.5 Main fund controls
- 2.6 Descriptive characteristics of concentrated funds

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Carhart four-factor performance model
- 3.2 Conditional Ferson-Schadt performance model
- 3.3 DGTW holding-based measures
- 3.4 Industry-adjusted measures
- 3.5 Trade portfolio measures
- 3.6 Panel regression specification
- 3.7 Portfolio-sorting design

4 Identification and instruments (what makes the estimates credible)

- 4.1 What identifies the main relation
- 4.2 Risk and style adjustment
- 4.3 Industry timing versus stock selection
- 4.4 Fund size and diseconomies of scale
- 4.5 Investment style
- 4.6 Trade-level validation
- 4.7 Remaining limitations

5 Tables and figures

- 5.1 Table I: Summary statistics
- 5.2 Table II: Decile portfolios, factor-based performance
- 5.3 Table III: Decile portfolios, holding-based performance
- 5.4 Table IV: Panel regression evidence
- 5.5 Table V: Alternative risk adjustments
- 5.6 Table VII: Size portfolios

5.7 Table VIII: Style portfolios

5.8 Table IX: Trade portfolios

6 Short summary

7 Conclusion

7.1 Core conclusion (what the paper establishes)

7.2 Economic intuition

7.3 Takeaway

1 Big picture

Question. Do actively managed equity mutual funds perform better when they concentrate their holdings in a few industries, rather than holding portfolios that look close to the broad market portfolio?

Setting. The paper studies U.S. actively managed diversified equity mutual funds from 1984 to 1999. The data combine CRSP mutual fund information, CDA/Spectrum quarterly holdings, and CRSP stock returns and classifications.

Core challenge. Concentration can mean two very different things. It may indicate that a manager has industry-specific information and deliberately focuses on industries where the manager has an advantage. But it may also indicate risk-taking or “gaming,” because concentrated portfolios are volatile and may help a weak manager create a chance of extreme performance.

Main conceptual contribution. The paper builds an *Industry Concentration Index* to measure how far a fund’s industry weights deviate from the market’s industry weights. The index is essentially a market-adjusted Herfindahl measure over 10 broad industries.

Main empirical design. The authors sort funds each quarter by lagged industry concentration and compare the performance of diversified versus concentrated funds. They also estimate panel regressions of fund performance on lagged concentration, controlling for expenses, turnover, age, size, new money growth, and time effects.

Main baseline result. More concentrated funds outperform more diversified funds. In portfolio tests, the most concentrated deciles have higher four-factor abnormal returns than the most diversified deciles. In panel regressions, a 5 percentage point increase in the concentration index, roughly one standard deviation, raises quarterly four-factor abnormal return by about 13 basis points, or roughly 0.52 percentage points per year before expenses.

Key mechanism result. The performance advantage is not only an industry-timing effect. Holding-based DGTW measures show that concentrated funds have stronger stock selection ability. Industry-adjusted measures also show that concentrated funds select better stocks *within* industries and, to a lesser extent, select better industries.

Key robustness result. The positive concentration–performance relation survives conditional factor models, holding-based benchmarks, industry adjustments, appraisal-ratio scaling, subperiod splits, fund-size splits, and investment-style splits.

Trade-level result. Concentrated funds’ purchases outperform their sales by more than diversified funds’ purchases outperform their sales. This is especially important because trade-based evidence is closer to the actual active decisions made by managers.

Economic takeaway. The evidence supports the view that investment ability is more visible among managers who concentrate in industries where they may have informational advantages. The paper does not say that all concentration is good; it says that, in this sample, concentration is systematically associated with superior active management.

2 Data and measurement (key objects)

2.1 Data sources and sample construction

The main data set merges three sources:

- the CRSP Survivorship Bias Free Mutual Fund Database, which provides fund returns, total net assets, fees, turnover, investment objectives, and other fund characteristics;
- the CDA/Spectrum holdings database, which provides mutual fund stock holdings;
- CRSP stock data, which provide stock returns, prices, and industry classification information.

The final sample covers January 1984 to December 1999 and contains 1,771 actively managed diversified U.S. equity funds.

Main exclusions. The authors remove balanced, bond, index, international, and sector funds. For funds with multiple share classes, they keep the fund only once. They also drop observations with fewer than 11 identified stock holdings and observations where the fund’s lagged total net assets do not exceed \$1 million.

Interpretation. The sample is designed to study broad active equity funds, not sector funds whose mandate mechanically requires industry concentration.

2.2 Industry groups

Each stock held by a fund is assigned to one of 10 broad industry groups. These 10 groups are constructed by aggregating the 48 Fama-French industry classifications. The broad groups include consumer non-durables, consumer durables, healthcare, manufacturing, energy, utilities, telecom, business equipment and services, wholesale and retail, and finance.

Interpretation. The paper measures whether a diversified equity fund deviates across broad economic sectors. This is different from asking whether a fund holds only a few individual stocks.

2.3 Industry Concentration Index

Let $w_{j,t}$ be the weight of a mutual fund in industry j at time t , and let $\bar{w}_{j,t}$ be the weight of the total stock market in industry j at time t . The Industry Concentration Index is

$$ICI_t = \sum_{j=1}^{10} (w_{j,t} - \bar{w}_{j,t})^2. \quad (1)$$

Interpretation. A fund has $ICI_t = 0$ if its industry composition exactly matches the market portfolio. The index rises when the fund overweights some industries and underweights others.

The index is related to the Herfindahl Index,

$$HI_t = \sum_{j=1}^{10} w_{j,t}^2, \quad (2)$$

but it adjusts for market-wide industry weights. This matters because the market itself is not equally weighted across industries.

Descriptive fact. The average fund has an Industry Concentration Index of 5.98%, with a median of 4.36%. The index ranges from 0.01% to 83.42%, so there is substantial cross-sectional variation in concentration.

2.4 Fund performance measures

The paper uses both return-based and holdings-based measures.

Before-expense performance. This is useful for evaluating investment skill because a skilled manager may charge higher fees.

After-expense performance. This is useful for evaluating what investors actually receive after paying fund expenses.

Factor-based alpha. The main return-based measure is the Carhart four-factor abnormal return, which controls for market, size, value, and momentum exposure.

Conditional alpha. The paper also uses a Ferson-Schadt conditional performance model, which allows the market beta to vary with macroeconomic state variables.

Holding-based performance. The DGTW measures decompose portfolio performance into stock selectivity, style timing, and average style.

Trade-based performance. The trade portfolio tests compare the future returns of stocks bought by funds with the future returns of stocks sold by funds.

2.5 Main fund controls

The regression controls include:

- EXP: expense ratio;
- TU: turnover ratio;
- LAGE: log fund age;
- LTNA: log total net assets;
- NMG: new money growth, a cash-flow measure for investor inflows;
- time dummies.

Interpretation. These controls address the concern that concentrated funds may differ from diversified funds in size, age, trading intensity, fees, or recent investor demand.

2.6 Descriptive characteristics of concentrated funds

Concentrated funds tend to have higher turnover and higher expenses. They are also younger and smaller on average. Their portfolios tend to tilt more toward small stocks and growth stocks than diversified funds.

Interpretation. Concentration is not a neutral portfolio feature. It is correlated with fund style and fund organization, so the empirical analysis must adjust for risk, style, and fund characteristics.

3 Models and empirical specifications (all equations + interpretation)

3.1 Carhart four-factor performance model

The baseline factor model is

$$R_{i,t} - R_{F,t} = \alpha_i + \beta_{i,M}(R_{M,t} - R_{F,t}) + \beta_{i,SMB}SMB_t + \beta_{i,HML}HML_t + \beta_{i,MOM}MOM_t + e_{i,t}. \quad (3)$$

Here $R_{i,t}$ is the return on fund portfolio i , $R_{F,t}$ is the risk-free rate, $R_{M,t} - R_{F,t}$ is the market excess

return, SMB_t is the size factor, HML_t is the value factor, and MOM_t is the momentum factor.

Interpretation. The intercept α_i is the fund's abnormal return after controlling for standard equity risk and style exposures. Momentum is especially important because industry momentum can be strong.

3.2 Conditional Ferson-Schadt performance model

The conditional model is

$$R_{i,t} - R_{F,t} = \alpha_i + \beta_{i,M}(R_{M,t} - R_{F,t}) + \beta_{i,SMB}SMB_t + \beta_{i,HML}HML_t + \beta_{i,MOM}MOM_t + \sum_{j=1}^4 \beta_{i,j} [z_{j,t-1}(R_{M,t} - R_{F,t})] + e_{i,t}. \quad (4)$$

The variables $z_{j,t-1}$ are lagged, demeaned macroeconomic variables: the one-month Treasury bill yield, the S&P 500 dividend yield, the Treasury term spread, and the corporate bond quality spread.

Interpretation. This model asks whether concentrated funds look good merely because they adjust market exposure in response to public macroeconomic conditions. The paper finds that the concentration result remains under this conditional model.

3.3 DGTW holding-based measures

The characteristic selectivity measure is

$$CS_t = \sum_j w_{j,t-1} [R_{j,t} - BR_t(j, t-1)]. \quad (5)$$

Here $R_{j,t}$ is the return on stock j , and $BR_t(j, t-1)$ is the return of a benchmark portfolio matched to stock j by size, book-to-market, and momentum at $t-1$.

Interpretation. CS_t measures stock-picking ability: did the stocks held by the fund outperform comparable stocks with similar characteristics?

The characteristic timing measure is

$$CT_t = \sum_j [w_{j,t-1}BR_t(j, t-1) - w_{j,t-5}BR_t(j, t-5)]. \quad (6)$$

Interpretation. CT_t measures whether the fund moves across styles at useful times.

The average style measure is

$$AS_t = \sum_j w_{j,t-5}BR_t(j, t-5). \quad (7)$$

Interpretation. AS_t captures the return earned from a fund's general tendency to hold certain kinds of stocks.

3.4 Industry-adjusted measures

To separate within-industry stock picking from industry choice, the paper defines

$$IS_t = \sum_j w_{j,t-1} [R_{j,t} - IR_t(j, t-1)], \quad (8)$$

where $IR_t(j, t-1)$ is the return on the industry portfolio to which stock j belonged at $t-1$.

Interpretation. IS_t measures stock selection within industries. It asks whether the manager picked better stocks than the industry benchmark.

The industry timing measure is

$$IT_t = \sum_j [w_{j,t-1} IR_t(j, t-1) - w_{j,t-5} IR_t(j, t-5)]. \quad (9)$$

Interpretation. IT_t measures whether the manager shifted capital across industries in a value-creating way.

3.5 Trade portfolio measures

For each fund, the paper compares the stocks bought and sold over the previous six months. The return on buys is

$$R_t^{\text{Buys}} = \frac{\sum_{w_{j,t-1} > \tilde{w}_{j,t-3}} (w_{j,t-1} - \tilde{w}_{j,t-3}) R_{j,t}}{\sum_{w_{j,t-1} > \tilde{w}_{j,t-3}} (w_{j,t-1} - \tilde{w}_{j,t-3})}. \quad (10)$$

The return on sells is

$$R_t^{\text{Sells}} = \frac{\sum_{w_{j,t-1} < \tilde{w}_{j,t-3}} (w_{j,t-1} - \tilde{w}_{j,t-3}) R_{j,t}}{\sum_{w_{j,t-1} < \tilde{w}_{j,t-3}} (w_{j,t-1} - \tilde{w}_{j,t-3})}. \quad (11)$$

The buy-and-hold adjusted lagged weight is

$$\tilde{w}_{j,t-3} = \frac{w_{j,t-3}(1 + R_{j,t-2})(1 + R_{j,t-1})}{\sum_j w_{j,t-3}(1 + R_{j,t-2})(1 + R_{j,t-1})}. \quad (12)$$

The buy-minus-sell return is

$$R_t^{\text{Buys-Sells}} = R_t^{\text{Buys}} - R_t^{\text{Sells}}. \quad (13)$$

Interpretation. This design controls for passive weight changes caused by price movements. A stock is classified as bought only if its portfolio weight rises more than it would have risen mechanically under buy-and-hold.

3.6 Panel regression specification

The main regression is

$$\text{PERF}_{i,t} = \beta_0 + \beta_1 \text{ICL}_{i,t-1} + \beta_2 \text{EXP}_{i,t-1} + \beta_3 \text{TU}_{i,t-1} + \beta_4 \text{LAGE}_{i,t-1} + \beta_5 \text{LTNA}_{i,t-1} + \beta_6 \text{NMG}_{i,t-1} + \gamma_t + \varepsilon_{i,t}. \quad (14)$$

The dependent variable $\text{PERF}_{i,t}$ is alternately four-factor abnormal return, conditional abnormal return, CS, CT, IS, IT, or the appraisal ratio. The regression includes time dummies γ_t and uses panel-corrected standard errors.

Interpretation. The coefficient of interest is β_1 . A positive β_1 means that funds with higher lagged industry concentration subsequently perform better, after controlling for observable fund characteristics and common time shocks.

3.7 Portfolio-sorting design

Each quarter, funds are sorted into deciles based on lagged ICI. The paper then computes equally weighted portfolio returns and performance measures for each concentration decile.

Interpretation. Sorting makes the pattern transparent: it shows whether performance rises monotonically from diversified to concentrated funds, rather than relying only on a regression coefficient.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main relation

There is no randomized treatment and no instrumental variable. The paper identifies a predictive relation between lagged industry concentration and subsequent fund performance by comparing funds with different concentration levels within the same time period.

Interpretation. The paper should be read as evidence that concentration is associated with skill, not as proof that mechanically forcing a manager to concentrate would improve performance.

4.2 Risk and style adjustment

A major concern is that concentrated funds may simply load on rewarded risk factors. The paper addresses this with the Carhart four-factor model and a conditional factor model. It also uses DGTW characteristic benchmarks, which compare each stock to stocks with similar size, book-to-market, and momentum characteristics.

Economic message. The result is not just that concentrated funds hold small stocks, growth stocks, or momentum stocks. The performance advantage survives direct controls for those characteristics.

4.3 Industry timing versus stock selection

Another concern is that concentrated funds may only benefit from being in hot industries. The paper separates industry effects using IS and IT.

Economic message. Concentrated funds outperform even when performance is adjusted for industry returns. This supports the interpretation that concentrated managers pick better stocks inside industries, rather than merely allocating to winning industries.

4.4 Fund size and diseconomies of scale

A separate literature shows that large funds may underperform because of diseconomies of scale. The paper therefore sorts funds by size and then compares high-concentration and low-concentration funds within size groups.

Economic message. The concentration result is not driven only by tiny funds. Concentrated funds tend to do better than diversified funds across fund-size quintiles.

4.5 Investment style

Concentrated funds are more likely to be small-growth oriented. The paper therefore sorts funds into small-growth, small-value, large-growth, and large-value styles, then compares high and low concentration funds within style.

Economic message. The concentration effect is strongest in growth-oriented styles, but the general evidence indicates that concentration is not merely a disguised style label.

4.6 Trade-level validation

Holdings can be noisy because a fund may hold inherited positions, passive positions, or positions that reflect past choices. The trade tests focus more directly on active buy and sell decisions.

Economic message. Concentrated funds' buys outperform their sells by more than diversified funds' buys outperform their sells. This is strong supporting evidence for active selection skill.

4.7 Remaining limitations

The paper cannot fully rule out all omitted manager characteristics. A concentrated fund may have better research technology, better incentives, or a different investor clientele. The evidence is therefore strongest as a test of whether concentration reveals skill, not as a full structural explanation of where skill comes from.

5 Tables and figures

5.1 Table I: Summary statistics

Read.

- The final sample includes 1,771 actively managed equity funds.
- The average fund holds about 97 stocks, while the median fund holds 65 stocks.
- The average Industry Concentration Index is 5.98%, with a median of 4.36%.
- Concentration is positively correlated with expenses and turnover, and negatively correlated with age and fund size.

Economic message. Concentrated funds are not identical to diversified funds. They are smaller, younger, more expensive, and more actively traded, so later tests need to control for these characteristics.

Table I
Summary Statistics

Panel A presents the summary statistics of the actively managed equity mutual funds included in the paper. Panel B reports the contemporaneous correlations between the main variables used in the paper. The Industry Concentration Index is defined as $ICI = \sum (w_j - \bar{w}_j)^2$, where w_j is the weight of the mutual fund holdings in industry j and $\bar{w}_j$ is the weight of the market in industry j .

Panel A: Fund Characteristics				
	Mean	Median	Minimum	Maximum
Total number of funds	1,771			
Number of stocks held by fund	97.12	65	11	3,439
TNA (total net assets) (in millions)	623.44	107.18	1.001	97,594
Age (years)	14.58	8	1	77
Expenses (%)	1.26	1.17	0.01	14.54
Turnover (%)	88.28	64.0	0.04	4,263
Total load (%)	2.55	0	0	8.98
Quarterly raw return (%)	4.44	4.29	-49.32	130.62
Industry Concentration Index (%)	5.98	4.36	0.01	83.42

Panel B: Correlation Structure						
Variables	Industry Concentration		Expenses	Turnover	Age	TNA
	Index	Loads				
Concentration Index	1.00					
Expenses	0.21***		1.00			
Turnover	0.15***		0.14***	1.00		
Age	-0.08***		-0.19***	-0.07***	1.00	
TNA	-0.06***		-0.15***	-0.03***	0.20***	1.00
Loads	-0.05***		0.01***	-0.04***	0.17***	0.02***

***1% significance, **5% significance, *10% significance.

Figure 1: Summary statistics and correlation structure.

5.2 Table II: Decile portfolios, factor-based performance

Read.

- Funds are sorted into deciles by lagged industry concentration.
- Concentrated funds have higher abnormal returns before expenses than diversified funds.
- The concentration ranking is also visible after expenses, though expenses absorb part of the value created by active managers.
- Concentrated funds have stronger small-stock, growth-stock, and momentum tilts, which motivates the use of factor controls.

Economic message. The basic portfolio evidence supports the central hypothesis: industry concentration is associated with superior risk-adjusted performance.

Table II
Decile Portfolios: Factor-Based Performance Measures

This table summarizes abnormal returns and the factor loadings using the Carhart (1997) four-factor model for different portfolios of mutual funds for the period of 1984 to 1999. The first and third columns show the unconditional abnormal returns before and after expenses. The second and fourth columns show the conditional abnormal returns according to Ferson and Schadt (1996), using the lagged level of the 1-month Treasury bill yield, the lagged dividend yield of the S&P 500 Index, the lagged measure of the slope of the term structure, and the lagged quality spread in the bond market. The last four columns summarize the factor loadings for the unconditional model using returns before expenses. We divide the sample into deciles based on the lagged Industry Concentration Index $ICI = \sum_i w_j - b_j^2$, where w_j is the weight of the mutual fund holdings in industry j and b_j is the weight of the market in industry j . The returns are expressed at a quarterly frequency and the portfolios are rebalanced quarterly. The standard errors of the regressions are given in parentheses. The table includes the differences in the abnormal returns along with their standard errors before and after expenses between the top and the bottom deciles, the top and the bottom quintiles, and the top and the bottom halves of the mutual funds. Spearman rank correlations have been included together with their respective p -values.

	Abnormal Return (% per quarter)				Factor Loadings Before Expenses Unconditional Model			
	Before Expenses		After Expenses		Market	Size	Value	Momentum
	Unconditional	Conditional	Unconditional	Conditional				
All funds	0.24*** (0.09)	0.21** (0.09)	-0.07 (0.09)	-0.11 (0.09)	0.96*** (0.01)	0.28*** (0.02)	-0.09*** (0.02)	0.03** (0.01)
Decile 1 (Diversified)	0.09 (0.10)	0.01 (0.10)	-0.17* (0.10)	-0.25** (0.10)	0.97*** (0.01)	-0.07*** (0.02)	0.01 (0.02)	0.00 (0.02)
Decile 2	0.08 (0.11)	0.03 (0.12)	-0.19* (0.11)	-0.24** (0.12)	0.96*** (0.01)	0.03 (0.02)	0.01 (0.02)	-0.01 (0.02)
Decile 3	0.10 (0.11)	0.07 (0.12)	-0.18 (0.11)	-0.22* (0.12)	0.96*** (0.01)	0.13*** (0.02)	0.01 (0.02)	0.00 (0.02)
Decile 4	0.08 (0.17)	-0.03 (0.17)	-0.21 (0.17)	-0.32* (0.17)	0.97*** (0.02)	0.21*** (0.03)	0.02 (0.03)	0.02 (0.03)
Decile 5	0.10 (0.17)	0.02 (0.17)	-0.21 (0.17)	-0.30* (0.17)	0.97*** (0.02)	0.24*** (0.03)	-0.00 (0.03)	-0.01 (0.02)
Decile 6	0.33* (0.18)	0.29 (0.19)	0.16 (0.18)	-0.03 (0.19)	0.97*** (0.02)	0.31*** (0.03)	-0.02 (0.03)	0.02 (0.03)
Decile 7	0.44** (0.19)	0.37* (0.20)	0.11 (0.19)	0.04 (0.20)	0.96*** (0.02)	0.38*** (0.04)	-0.07** (0.04)	0.03 (0.03)

(continued)

Figure 2: Factor-based performance by concentration decile, part A.

Table II—Continued

	Abnormal Return (% per quarter)				Factor Loadings Before Expenses Unconditional Model			
	Before Expenses		After Expenses		Market	Size	Value	Momentum
	Unconditional	Conditional	Unconditional	Conditional				
Decile 8	0.26* (0.16)	0.30* (0.16)	-0.08 (0.16)	-0.04 (0.16)	0.97*** (0.02)	0.49*** (0.03)	-0.16*** (0.03)	0.05** (0.02)
Decile 9	0.42** (0.18)	0.41** (0.20)	0.07 (0.18)	0.07 (0.20)	0.99*** (0.02)	0.49*** (0.04)	-0.25*** (0.03)	0.07** (0.03)
Decile 10 (Concentrated)	0.53* (0.29)	0.59** (0.30)	0.15 (0.29)	0.22 (0.30)	0.93*** (0.04)	0.64*** (0.05)	-0.47*** (0.05)	0.12*** (0.04)
2 nd half–1 st half	0.30** (0.14)	0.37*** (0.14)	0.24* (0.14)	0.32** (0.14)	-0.00 (0.02)	0.35*** (0.03)	-0.20*** (0.03)	0.06*** (0.02)
5 th quintile–1 st quintile	0.59** (0.20)	0.48** (0.21)	0.29 (0.20)	0.39* (0.21)	-0.01 (0.03)	0.58*** (0.04)	-0.37*** (0.04)	0.10*** (0.03)
10 th decile–1 st decile	0.44 (0.30)	0.58* (0.32)	0.32 (0.30)	0.47 (0.32)	-0.04 (0.04)	0.70*** (0.06)	-0.48*** (0.06)	0.12** (0.05)
Spearman rank correlation	0.87*** (0.00)	0.85*** (0.00)	0.71** (0.02)	0.78*** (0.01)	-0.04 (0.91)	0.99*** (0.00)	-0.92*** (0.00)	0.82*** (0.00)

***1% significance, **5% significance, *10% significance.

Figure 3: Factor-based performance by concentration decile, part B.

5.3 Table III: Decile portfolios, holding-based performance

Read.

- The characteristic selectivity measure CS rises with concentration.
- The characteristic timing measure CT also has a positive rank relation with concentration.
- Average style AS does not explain the performance advantage of concentrated funds.

Economic message. The outperformance is tied to active holdings and stock selection, not only to factor-model estimation.

Table III
Decile Portfolios: Holding-Based Performance Measures

This table summarizes holding-based performance measures according to DGTW (1997) for different portfolios of mutual funds for the period of 1984 to 1999. We divide the sample into deciles based on the lagged Industry Concentration Index, which is defined as $ICI = \sum (w_j - \bar{w}_j)^2$, where w_j is the weight of the mutual fund holdings in industry j and $\bar{w}_j$ is the weight of the market in industry j . The returns are expressed at a quarterly frequency and the portfolios are rebalanced quarterly. The characteristic-based performance measures are denoted by CS, CT, and AS. The stock selection ability is defined as $CS = \sum w_{j,t-1} [R_{j,t} - BR_t(j, t-1)]$, where $BR_t(j, t-1)$ denotes the return of a benchmark portfolio during period t to which stock j was allocated during period $t-1$ according to its size, value, and momentum characteristics. The style-timing ability is defined as $CT = \sum [w_{j,t-1} BR_t(j, t-1) - w_{j,t-5} BR_t(j, t-5)]$ and the style-selection ability is defined as $AS = \sum [w_{j,t-1} BR_t(j, t-1) - w_{j,t-5} BR_t(j, t-5)]$. The standard errors of the regressions are given in parentheses. The table includes the differences in the abnormal returns along with their standard errors between the top and the bottom deciles, the top and the bottom quintiles, and the top and the bottom halves of the mutual funds. Spearman rank correlations have been included together with their respective p -values.

Deciles	Holding-Based Performance (% per quarter)		
	CS	CT	AS
All funds	0.24* (0.13)	0.08 (0.06)	4.26*** (1.12)
Decile 1 (Diversified)	0.13 (0.12)	0.03 (0.08)	4.48*** (1.03)
Decile 2	0.14 (0.11)	0.04 (0.07)	4.36*** (1.04)
Decile 3	0.13 (0.11)	0.04 (0.06)	4.31*** (1.07)
Decile 4	0.16 (0.12)	0.08 (0.06)	4.23*** (1.08)
Decile 5	0.14 (0.12)	0.06 (0.06)	4.20*** (1.09)
Decile 6	0.24* (0.13)	0.05 (0.06)	4.26*** (1.12)
Decile 7	0.33** (0.17)	0.09 (0.06)	4.17*** (1.14)
Decile 8	0.21 (0.20)	0.11 (0.08)	4.17*** (1.18)
Decile 9	0.40* (0.24)	0.15* (0.09)	4.17*** (1.20)
Decile 10	0.53 (0.33)	0.13 (0.10)	4.22*** (1.27)
2 nd half–1 st half	0.20 (0.15)	0.06 (0.05)	–0.12 (0.20)
5 th quintile–1 st quintile	0.33 (0.28)	0.11 (0.08)	–0.23 (0.34)
10 th decile–1 st decile	0.40 (0.34)	0.11 (0.11)	–0.26 (0.41)
Spearman rank correlation	0.88*** (0.00)	0.93*** (0.00)	–0.82*** (0.00)

***1% significance, **5% significance, *10% significance.

Figure 4: Holding-based DGTW performance measures.

5.4 Table IV: Panel regression evidence

Read.

- The coefficient on ICI is positive and significant for unconditional four-factor alpha.
- The coefficient remains positive and significant for conditional alpha.
- ICI is also positive and significant for CS and CT.
- A 5 percentage point increase in ICI implies about 13 basis points more quarterly four-factor abnormal return.

Economic message. The concentration effect survives fund-level controls and time effects. The result is not just a decile-sorting artifact.

Table IV
Regression Evidence

This table reports the coefficients of the quarterly panel regression of the general form: $PERF_{i,t} = \beta_0 + \beta_1 \times ICI_{i,t-1} + \beta_2 \times EXP_{i,t-1} + \beta_3 \times TU_{i,t-1} + \beta_4 \times LAGE_{i,t-1} + \beta_5 \times LTNA_{i,t-1} + \beta_6 \times NMG_{i,t-1} + \varepsilon_{i,t}$. The sample includes actively managed equity mutual funds and spans the period of 1984 to 1999 (including the data used for calculating the abnormal returns). The dependent variable, PERF, measures the quarterly performance using the four-factor model of Carhart (1997) based on 36 months of lagged data, the conditional performance according to Ferson and Schadt (1996), and the holding-based performance measures, CS and CT according to DGTW (1997). The Industry Concentration Index is defined as $ICI = \sum (w_j - \bar{w}_j)^2$, where w_j is the weight of the mutual fund holdings in industry j and $\bar{w}_j$ is the weight of the market in industry j . We denote the expense ratio by EXP, the turnover ratio by TU, the natural logarithm of age by LAGE, the natural logarithm of total net assets by LTNA, and the new money growth by NMG. All regressions include time dummies. Panel-corrected standard errors are reported in parentheses.

	Dependent Variable: Quarterly Performance (bp)			
	Four-Factor Abnormal Return		Holding-Based Performance	
	Unconditional	Conditional	CS	CT
ICI (%)	2.57*** (0.62)	2.82*** (0.77)	2.94*** (0.56)	0.94*** (0.23)
EXP (%)	-40.16*** (6.84)	-46.07*** (10.26)	-2.15 (5.70)	2.61 (2.48)
TU (%)	0.04 (0.05)	0.07 (0.06)	0.18*** (0.04)	0.06*** (0.02)
LAGE	-12.37*** (3.21)	-20.10*** (4.18)	1.71 (2.70)	-1.66 (1.34)
LTNA	-1.16 (1.64)	-1.65 (2.09)	2.15 (1.53)	0.16 (0.74)
NMG	0.16 (0.18)	0.16 (0.23)	0.28** (0.13)	0.20*** (0.07)
Number of observations	30,645	30,645	42,659	36,325

***1% significance, **5% significance, *10% significance.

Figure 5: Regression evidence for factor-based and holding-based performance.

5.5 Table V: Alternative risk adjustments

Read.

- *ICI* is positive and significant for industry-adjusted stock selectivity IS.
- *ICI* is positive and significant for industry timing IT.
- The appraisal-ratio result shows that the performance advantage is not simply compensation for higher idiosyncratic volatility.

Economic message. Concentrated funds add value both by picking stocks within industries and by allocating across industries. The result is not driven only by riskier residual returns.

Table V
Regression Evidence: Alternative Risk Adjustments

This table reports the coefficients of the quarterly panel and cross-sectional regression of the general form: $PERF_{i,t} = \beta_0 + \beta_1 \times ICI_{i,t-1} + \beta_2 \times EXP_{i,t-1} + \beta_3 \times TU_{i,t-1} + \beta_4 \times LAGE_{i,t-1} + \beta_5 \times LTNA_{i,t-1} + \beta_6 \times NMG_{i,t-1} + \varepsilon_{i,t}$. The sample includes actively managed equity mutual funds and spans the period of 1984 to 1999 (including the data used for calculating the abnormal returns). The dependent variable, PERF, equals the industry-adjusted stock selectivity measure (IS), the industry-adjusted timing measure (IT), or the appraisal ratio of Treynor and Black (1973) based on the four-factor model. The Industry Concentration Index is defined as $ICI = \sum (w_j - \bar{w}_j)^2$, where w_j is the weight of the mutual fund holdings in industry j and $\bar{w}_j$ is the weight of the market in industry j . We denote the expense ratio by EXP, the turnover ratio by TU, the natural logarithm of age by LAGE, the natural logarithm of total net assets by LTNA, and the new money growth by NMG. All regressions include time dummies. Panel-corrected standard errors are reported in parentheses.

	Dependent Variable: Quarterly Performance (bp) Industry-Adjusted Abnormal Performance		Appraisal Ratio Four-Factor
	IS	IT	
ICI (%)	1.89*** (0.47)	1.46*** (0.19)	0.65*** (0.09)
EXP (%)	7.32 (5.98)	3.24 (2.78)	-6.58*** (0.95)
TU (%)	0.10*** (0.03)	0.12*** (0.02)	0.01** (0.01)
LAGE	-1.51 (2.32)	-8.72*** (1.15)	-2.54*** (0.72)
LTNA	9.91*** (1.38)	2.00*** (0.62)	-0.43 (0.35)
NMG	0.23** (0.11)	0.13** (0.06)	0.91 (3.49)
Number of observations	37,177	33,025	30,645

***1% significance, **5% significance, *10% significance.

Figure 6: Alternative risk adjustments and industry-adjusted measures.

5.6 Table VII: Size portfolios

Read.

- Funds are first sorted into size quintiles and then split into high- and low-concentration groups.
- Within each size group, high-concentration funds generally have better abnormal performance.
- The concentration effect is not limited to the smallest funds.

Economic message. The paper separates concentration from the known effect that small funds may perform better than large funds.

Table VII
Size Portfolios

Mutual funds are sorted into five equally sized portfolios according to the lagged TNA of the mutual funds. The mutual funds in each of these five portfolios are further divided into two groups according to the lagged Industry Concentration Index. The Industry Concentration Index is defined as $ICI = \sum (w_j - \bar{w}_j)^2$, where w_j is the weight of the mutual fund holdings in industry j and $\bar{w}_j$ is the weight of the market in industry j . The returns are expressed at a quarterly frequency and the portfolios are rebalanced quarterly. The abnormal returns before expenses using the Carhart (1997) four-factor model are summarized for different portfolios of mutual funds for the period of 1984 to 1999. The characteristic-based performance measures are denoted by CS and CT. The stock selection ability is defined as $CS = \sum w_{j,t-1} [R_{j,t} - BR_t(j, t-1)]$, where $BR_t(j, t-1)$ denotes the return of a benchmark portfolio during period t to which stock j was allocated during period $t-1$ according to its size, value, and momentum characteristics. The style-timing ability is defined as $CT = \sum [w_{j,t-1} BR_t(j, t-1) - w_{j,t-5} BR_t(j, t-5)]$. The standard errors of the regressions are given in parentheses.

Size Quintiles	Industry Concentration	Four-Factor Abnormal Return		Holding-Based Performance Measures	
		Unconditional	Conditional	CS	CT
Quintile 1	Low	0.36** (0.15)	0.28* (0.15)	0.19* (0.11)	0.08 (0.07)
		0.60*** (0.19)	0.56*** (0.20)	0.26 (0.20)	0.12 (0.10)
	High	0.24 (0.20)	0.28 (0.20)	0.07 (0.15)	0.04 (0.08)
Quintile 2	Low	0.08 (0.16)	0.00 (0.17)	0.13 (0.11)	0.08 (0.07)
		0.38** (0.15)	0.41*** (0.15)	0.43** (0.17)	0.14* (0.08)
	High	0.30** (0.15)	0.40*** (0.15)	0.30** (0.15)	0.05 (0.06)
Quintile 3	Low	0.07 (0.14)	-0.04 (0.14)	0.10 (0.11)	0.01 (0.06)
		0.27 (0.19)	0.27 (0.20)	0.28 (0.23)	0.09 (0.06)
	High	0.20 (0.19)	0.31* (0.20)	0.18 (0.20)	0.07 (0.05)
Quintile 4	Low	0.02 (0.14)	-0.05 (0.10)	0.14 (0.11)	0.03 (0.08)
		0.34* (0.20)	0.31 (0.21)	0.38* (0.20)	0.08 (0.07)
	High	0.33* (0.17)	0.35* (0.18)	0.24 (0.17)	0.05 (0.06)
Quintile 5	Low	0.07 (0.10)	0.05 (0.10)	0.16 (0.12)	0.05 (0.07)
		0.24 (0.18)	0.28 (0.19)	0.27 (0.21)	0.10 (0.07)
	High	0.18 (0.15)	0.23 (0.15)	0.11 (0.16)	0.05 (0.06)

***1% significance, **5% significance, *10% significance.

Figure 7: Performance by fund-size portfolio and concentration group.

5.7 Table VIII: Style portfolios

Read.

- Funds are grouped into small-growth, small-value, large-growth, and large-value styles.
- High-concentration funds outperform low-concentration funds most clearly in growth-oriented categories.
- The large-value category is the main place where the high-minus-low concentration difference is weak or negative.

Economic message. Concentration is related to style, but it is not reducible to style. The most compelling concentration effect appears where information advantages may be especially valuable:

growth stocks.

Table VIII
Style Portfolios

Mutual funds are sorted into four portfolios according to the lagged market values (small vs. large cap) and the lagged book-to-market ratios (growth vs. value) of their holdings. The mutual funds in each of these four portfolios are further divided into two groups according to the lagged Industry Concentration Index. The Industry Concentration Index is defined as $ICI = \sum (w_j - \bar{w}_j)^2$, where w_j is the weight of the mutual fund holdings in industry j and $\bar{w}_j$ is the weight of the market in industry j . The returns are expressed at a quarterly frequency and the portfolios are rebalanced quarterly. The abnormal returns before expenses using the Carhart (1997) four-factor model are summarized for different portfolios of mutual funds for the period of 1984 to 1999. The characteristic-based performance measures are denoted by CS and CT. The stock selection ability is defined as $CS = \sum w_{j,t-1} [R_{j,t} - BR_t(j, t-1)]$, where $BR_t(j, t-1)$ denotes the return of a benchmark portfolio during period t to which stock j was allocated during period $t-1$ according to its size, value, and momentum characteristics. The style-timing ability is defined as $CT = \sum [w_{j,t-1} BR_t(j, t-1) - w_{j,t-5} BR_t(j, t-5)]$. The standard errors of the regressions are given in parentheses.

Style	Industry Concentration	Four-Factor Abnormal Return		Holding-Based Performance Measures	
		Unconditional	Conditional	CS	CT
Small growth	Low	0.18 (0.21)	0.02 (0.20)	0.21 (0.16)	0.08 (0.08)
	High	0.59** (0.28)	0.72** (0.29)	0.62* (0.37)	0.19 (0.14)
	<i>High-Low</i>	0.40 (0.33)	0.70** (0.31)	0.41 (0.28)	0.11 (0.11)
Small value	Low	0.06 (0.20)	0.04 (0.22)	0.09 (0.13)	0.03 (0.05)
	High	0.41** (0.20)	0.41* (0.21)	0.14 (0.17)	0.08 (0.05)
	<i>High-Low</i>	0.35** (0.15)	0.37** (0.15)	0.05 (0.11)	0.05 (0.04)
Large growth	Low	0.12 (0.14)	-0.01 (0.13)	0.18 (0.13)	0.08 (0.09)
	High	0.41** (0.20)	0.39* (0.20)	0.41* (0.24)	0.16 (0.10)
	<i>High-Low</i>	0.29 (0.21)	0.41** (0.20)	0.24 (0.20)	0.07 (0.06)
Large value	Low	0.06 (0.17)	0.01 (0.18)	0.09 (0.13)	0.01 (0.09)
	High	-0.08 (0.20)	-0.14 (0.21)	0.06 (0.16)	-0.02 (0.09)
	<i>High-Low</i>	-0.14 (0.12)	-0.15 (0.12)	-0.03 (0.09)	-0.03 (0.05)

***1% significance, **5% significance, *10% significance.

portfolios. Finally, we subdivide each of these four portfolios according to their

Figure 8: Performance by investment style and concentration group.

5.8 Table IX: Trade portfolios

Read.

- For all funds, bought stocks outperform sold stocks.
- The buy-minus-sell gap increases strongly with industry concentration.
- In the most concentrated decile, the raw buy-minus-sell return is 2.11% per quarter, compared with 0.95% for the most diversified decile.
- The same pattern appears for style-adjusted CS returns.

Economic message. Concentrated funds make more valuable active trades. This supports the interpretation that concentration reflects information-based stock selection rather than only passive industry exposure.

Table IX
Trade Portfolios

This table summarizes the returns of the stocks purchased and sold by different portfolios of mutual funds for the period of 1984 to 1999. We divide the sample into deciles based on the lagged Industry Concentration Index, which is defined as $ICI = \sum (w_j - \bar{w}_j)^2$, where w_j is the weight of the mutual fund holdings in industry j and $\bar{w}_j$ is the weight of the market in industry j . The returns are expressed at a quarterly frequency and the portfolios are rebalanced quarterly. For each mutual fund, we compute the raw returns and style-adjusted returns of their stock purchases and sells. The style adjusted return is a measure of stock selection ability and is defined as $CS = \sum w_{j,t-1} [R_{j,t} - BR_t(j, t-1)]$, where $BR_t(j, t-1)$ is the return of a benchmark portfolio during period t to which stock j was allocated during period $t-1$ according to its size, value, and momentum characteristics. The table includes the differences in the returns along with their standard errors between the top and the bottom deciles, the top and the bottom quintiles, and the top and the bottom halves of the mutual funds. Spearman rank correlations have been included together with their respective p -values.

	Raw Returns			CS Measure		
	Buys	Sells	Buys-Sells	Buys	Sells	Buys-Sells
All funds	5.01*** (1.31)	3.66*** (1.22)	1.35*** (0.42)	0.57*** (0.22)	-0.50* (0.29)	1.06*** (0.31)
Decile 1 (Diversified)	4.96*** (1.12)	4.01*** (1.09)	0.95** (0.40)	0.33** (0.16)	-0.28 (0.26)	0.61** (0.28)
Decile 2	4.99*** (1.18)	3.91*** (1.09)	1.08*** (0.41)	0.45*** (0.16)	-0.26 (0.25)	0.71** (0.28)
Decile 3	4.81*** (1.19)	3.68*** (1.16)	1.13*** (0.37)	0.39** (0.16)	-0.52* (0.27)	0.92*** (0.28)
Decile 4	4.93*** (1.24)	3.70*** (1.19)	1.24*** (0.39)	0.50*** (0.18)	-0.46 (0.29)	0.97*** (0.30)
Decile 5	4.79*** (1.26)	3.61*** (1.20)	1.17*** (0.41)	0.41** (0.19)	-0.52 (0.31)	0.93*** (0.35)
Decile 6	4.86*** (1.30)	3.73*** (1.24)	1.13** (0.36)	0.45** (0.20)	-0.43 (0.32)	0.88*** (0.29)
Decile 7	4.91*** (1.34)	3.50*** (1.25)	1.40*** (0.41)	0.54** (0.23)	-0.62* (0.33)	1.16*** (0.33)
Decile 8	5.00*** (1.46)	3.36** (1.32)	1.64*** (0.50)	0.61* (0.34)	-0.72* (0.38)	1.32*** (0.39)
Decile 9	5.20*** (1.51)	3.53*** (1.33)	1.66*** (0.52)	0.76** (0.37)	-0.60 (0.39)	1.35*** (0.38)
Decile 10 (Concentrated)	5.69*** (1.68)	3.57** (1.47)	2.11*** (0.62)	1.28** (0.52)	-0.55 (0.48)	1.82*** (0.49)
2 nd half-1 st half	0.23 (0.37)	-0.24 (0.30)	0.48*** (0.17)	0.31 (0.22)	-0.17 (0.23)	0.48*** (0.17)
5 th quintile-1 st quintile	0.46 (0.64)	-0.41 (0.52)	0.87*** (0.30)	0.62 (0.40)	-0.31 (0.39)	0.93*** (0.29)
10 th decile-1 st decile	0.72 (0.81)	-0.44 (0.64)	1.16*** (0.40)	0.94* (0.52)	-0.27 (0.48)	1.21*** (0.39)
Spearman rank correlation	0.48 (0.16)	-0.81*** (0.00)	0.94*** (0.00)	0.89*** (0.00)	-0.78*** (0.01)	0.95*** (0.00)

***1% significance, **5% significance, *10% significance.

Figure 9: Returns on stocks bought and sold by concentration decile.

6 Short summary

- The paper asks whether industry concentration reveals mutual fund manager skill.
- It constructs an Industry Concentration Index measuring how far a fund's industry weights deviate from the market's industry weights.
- Concentrated funds outperform diversified funds using four-factor abnormal returns, conditional alphas, holding-based measures, industry-adjusted measures, appraisal ratios, and trade-based measures.
- The evidence is strongest for stock selection: concentrated funds select better stocks, including better stocks within industries.

- The performance advantage survives controls for fund expenses, turnover, age, size, new money growth, and time effects.
- The paper's interpretation is that skilled managers concentrate where they have informational advantages.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

Kacperczyk, Sialm, and Zheng show that actively managed equity mutual funds with higher industry concentration tend to perform better than more diversified funds. The result holds across multiple performance measures and empirical designs.

7.2 Economic intuition

Diversification reduces idiosyncratic risk, so a fund manager should normally hold a broadly diversified portfolio. But a manager with superior information about specific industries has a reason to deviate from the market portfolio. Concentration is therefore potentially a signal of specialized information.

The paper's evidence favors the specialized-information interpretation over the pure gaming interpretation. If concentration were mainly risk-taking by weak managers, concentrated funds should not systematically outperform after risk and style adjustment. Instead, the paper finds that concentrated funds pick better stocks and make better trades.

7.3 Takeaway

Industry concentration is not automatically bad diversification. For active mutual funds, it can be an empirical marker of where managerial skill is most visible. The paper's main contribution is to show that the value of active management may be easier to detect among managers who make concentrated, information-based portfolio choices.